```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
int main(){
   FILE *fp = fopen("input.c", "r");
   char ch, str[30], keyword[][10] =
{"int","char","if","else",

"while","for","return","void"};
   int keycount = 8, line = 1, token = 0, i, j,
flag;

printf("Line\tToken\tType\t\tLexeme\n");
   while((ch = fgetc(fp)) != EOF){
      if(ch == '\n'){ line++; continue; }
      if(ch == ' ' || ch == '\t') continue;
      if(isalpha(ch)){
         i = 0;
         str[i++] = ch;
         while(isalnum(ch = fgetc(fp)))
str[i++] = ch;
         str[i] = '\0';
         flag = 0;
         for(j = 0; j < keycount; j++)
            if(strcmp(str, keyword[j]) == 0)
flag = 1;
         if(flag){

printf("%d\t%d\tKeyword\t\t%s\n", line,
++token, str);
            ungetc(ch, fp);
         }
         else if(ch == '('){
            printf("%d\t%d\tFunction
Name\t%s\n", line, ++token, str);
            ungetc(ch, fp);
         }
         else{

printf("%d\t%d\tIdentifier\t%s\n", line,
++token, str);
            ungetc(ch, fp);
         }
         continue;
      }
      if(isdigit(ch)){
         i = 0;
         str[i++] = ch;
         while(isdigit(ch = fgetc(fp)))
str[i++] = ch;
         str[i] = '\0';
         ungetc(ch, fp);
         printf("%d\t%d\tNumeric
Literal\t%s\n", line, ++token, str);
         continue;
      }
      if(ch == '"'){
         i = 0;
         str[i++] = ch;
         while((ch = fgetc(fp)) != '"' && ch
!= EOF)
            str[i++] = ch;
         str[i++] = '"';
         str[i] = '\0';
      }
         printf("%d\t%d\tString
Literal\t%s\n", line, ++token, str);
         continue;
      if(strchr("+-*/=<>", ch)){
         printf("%d\t%d\tOperator\t%c\n",
line, ++token, ch);
         continue;
      }
      if(strchr("(){};,#", ch)){
         printf("%d\t%d\tSpecial
Symbol\t%c\n", line, ++token, ch);
         continue;
      }
   }
   fclose(fp);
   return 0;
}
```

```
%{
#include <stdio.h>
%}
%%
"int"|"char"|"if"|"else"|"while"|"for"|"return"|"void"
                                { printf("%s : Keyword\n", yytext); }
[a-zA-Z_][a-zA-Z0-9_]*"("      { yytext[yyleng-1] = '\0';
                            printf("%s : Function Name\n", yytext);
                            printf("( : Special Symbol\n"); }
[a-zA-Z_][a-zA-Z0-9_]*          { printf("%s : Identifier\n", yytext); }
\"([^\"\\]|\\.)*\"          { printf("%s : String Literal\n", yytext); }
[0-9]+                        { printf("%s : Numeric Literal\n", yytext); }
[+\-*/=<>]                     { printf("%s : Operator\n", yytext); }
[(){};,]                     { printf("%s : Special Symbol\n", yytext); }
[ \t\n]+                    { }
.                           { }
%%
int yywrap(){ return 1; }
int main(){
   yyin = fopen("input.c", "r");
   yylex();
   return 0;
}
```

```
%{
#include <stdio.h>
%}

%%
"abc"    { printf("ABC"); }
%%

int yywrap() {
   return 1;
}

int main() {
   printf("Enter the input string:\n");
   yylex();
   return 0;
}
```

```
%{
#include <stdio.h>

int lines = 0, words = 0, chars = 0;
%}

%%
\n          { lines++; chars++; }
[^ \t\n]+   { words++; chars += yyleng; }
.           { chars++; }
%%

int yywrap() {
   return 1;
}

int main() {
   printf("Enter text (Ctrl+D to end):\n");

   yylex();
   printf("Lines = %d\n", lines);
   printf("Words = %d\n", words);
   printf("Characters = %d\n", chars);

   return 0;
}
```

OUTPUT :

```
Enter the input string:
abc my name is ABC abcde
ABC my name is ABC ABCde
```

OUTPUT :

```
Enter text (Ctrl+D to end):
Hello! I am a computer

Lines = 1
Words = 5
Characters = 23
```

OUTPUT :

```
int       : Keyword
main      : Function Name
(         : Special Symbol
void      : Keyword
)         : Special Symbol
{         : Special Symbol
printf    : Function Name
(         : Special Symbol
"Hello, World!\n"       : String Literal
)         : Special Symbol
;         : Special Symbol
return    : Keyword
0         : Numeric Literal
;         : Special Symbol
}         : Special Symbol
```

OUTPUT :

Line	Token	Type	Lexeme
1	1	Keyword	int
1	2	Function Name	main
1	3	Special Symbol	(
1	4	Keyword	void
1	5	Special Symbol	)
1	6	Special Symbol	{
2	7	Function Name	printf
2	8	Special Symbol	(
2	9	String Literal	"Hello, World!\n"
2	10	Special Symbol	)
2	11	Special Symbol	;
3	12	Keyword	return
3	13	Numeric Literal	0
3	14	Special Symbol	;
4	15	Special Symbol	}